

Heat Transfer Through Air Between a Hot Source and Cold Wall

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February 3, 2026

1 Problem Statement

Consider heat transfer through an air gap between a hot surface at $T_h = 90^\circ\text{C}$ and a cold wall at $T_c = 20^\circ\text{C}$. The air gap has thickness $L = 0.05\text{ m}$ and cross-sectional area $A = 1\text{ m}^2$. We analyze two cases:

1. **Stagnant air** – heat transfer by conduction and natural convection
2. **Moving air** – heat transfer by forced convection

2 System Diagram

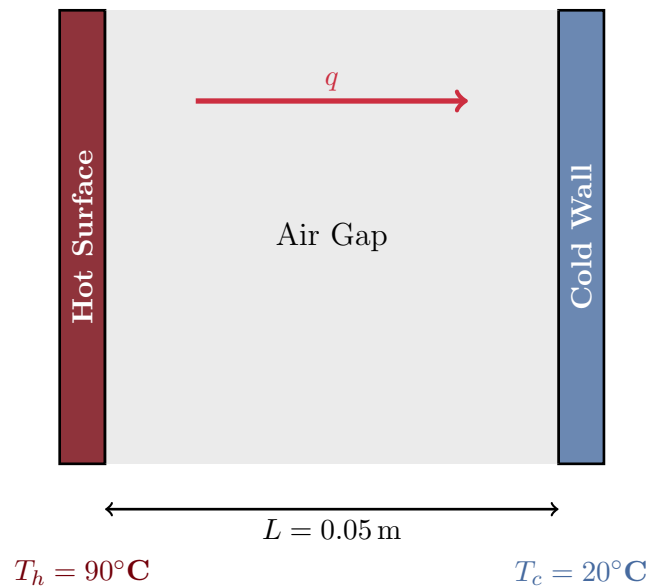


Figure 1: Schematic of heat transfer through an air gap between hot and cold surfaces.

3 Air Properties

We evaluate air properties at the film temperature:

$$T_f = \frac{T_h + T_c}{2} = \frac{90 + 20}{2} = 55^\circ\text{C} = 328\text{ K} \quad (1)$$

Table 1: Air properties at $T_f = 55^\circ\text{C}$

Property	Symbol	Value
Thermal conductivity	k	$0.0283\text{ W}/(\text{m} \cdot \text{K})$
Density	ρ	$1.076\text{ kg}/\text{m}^3$
Dynamic viscosity	μ	$1.96 \times 10^{-5}\text{ Pa} \cdot \text{s}$
Kinematic viscosity	ν	$1.82 \times 10^{-5}\text{ m}^2/\text{s}$
Specific heat	c_p	$1007\text{ J}/(\text{kg} \cdot \text{K})$
Thermal diffusivity	α	$2.59 \times 10^{-5}\text{ m}^2/\text{s}$
Prandtl number	Pr	0.703
Thermal expansion coeff.	β	$3.05 \times 10^{-3}\text{ K}^{-1}$

4 Case 1: Stagnant Air (Natural Convection)

4.1 Pure Conduction (Lower Bound)

If no convection occurs, heat transfers only by conduction:

$$q_{cond} = \frac{kA(T_h - T_c)}{L} = \frac{0.0283 \times 1 \times (90 - 20)}{0.05} = \boxed{39.6\text{ W}} \quad (2)$$

4.2 Natural Convection in Enclosure

For a vertical enclosure, we first calculate the Rayleigh number:

$$Ra_L = \frac{g\beta(T_h - T_c)L^3}{\nu\alpha} \quad (3)$$

$$Ra_L = \frac{9.81 \times 3.05 \times 10^{-3} \times 70 \times (0.05)^3}{1.82 \times 10^{-5} \times 2.59 \times 10^{-5}} = \boxed{5.58 \times 10^5} \quad (4)$$

For a vertical enclosure with aspect ratio $H/L = 20$ (assuming $H = 1\text{ m}$), we use the Berkovsky-Polevikov correlation:

$$Nu = 0.18 \left(\frac{Pr}{0.2 + Pr} Ra_L \right)^{0.29} \quad (5)$$

$$Nu = 0.18 \left(\frac{0.703}{0.2 + 0.703} \times 5.58 \times 10^5 \right)^{0.29} = \boxed{7.42} \quad (6)$$

The effective thermal conductivity and heat transfer rate:

$$k_{eff} = k \cdot Nu = 0.0283 \times 7.42 = 0.210\text{ W}/(\text{m} \cdot \text{K}) \quad (7)$$

$$q_{nat} = \frac{k_{eff}A(T_h - T_c)}{L} = \frac{0.210 \times 1 \times 70}{0.05} = \boxed{294\text{ W}} \quad (8)$$

The convective heat transfer coefficient for natural convection:

$$h_{nat} = \frac{Nu \cdot k}{L} = \frac{7.42 \times 0.0283}{0.05} = \boxed{4.2\text{ W}/(\text{m}^2 \cdot \text{K})} \quad (9)$$

5 Case 2: Moving Air (Forced Convection)

5.1 Flow Configuration

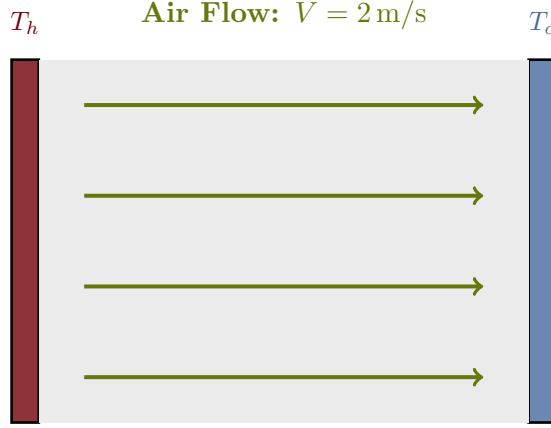


Figure 2: Forced convection with air flowing through the gap at velocity V .

Assume air flows through the gap at velocity $V = 2 \text{ m/s}$.

5.2 Reynolds Number

Using the gap width as the characteristic length:

$$Re_L = \frac{\rho V L}{\mu} = \frac{1.076 \times 2 \times 0.05}{1.96 \times 10^{-5}} = \boxed{5490} \quad (10)$$

This indicates **transitional flow** ($Re < 10^5$ for external flow over a plate).

5.3 Nusselt Number and Heat Transfer Coefficient

For flow between parallel plates (internal flow, developing), we use:

$$Nu = 7.54 + \frac{0.03(D_h/L_{flow})Re \cdot Pr}{1 + 0.016[(D_h/L_{flow})Re \cdot Pr]^{2/3}} \quad (11)$$

For a simpler estimate using flat plate correlation (laminar):

$$Nu = 0.664 Re_L^{0.5} Pr^{1/3} = 0.664 \times (5490)^{0.5} \times (0.703)^{1/3} = \boxed{43.7} \quad (12)$$

The forced convection heat transfer coefficient:

$$h_{forced} = \frac{Nu \cdot k}{L} = \frac{43.7 \times 0.0283}{0.05} = \boxed{24.7 \text{ W/(m}^2 \cdot \text{K)}} \quad (13)$$

Heat transfer rate:

$$q_{forced} = h_{forced} A (T_h - T_c) = 24.7 \times 1 \times 70 = \boxed{1729 \text{ W}} \quad (14)$$

Table 2: Comparison of heat transfer modes

Mode	h [$\text{W}/(\text{m}^2\text{K})$]	q [W]	Enhancement
Pure Conduction	0.57	39.6	$1.0\times$
Natural Convection	4.2	294	$7.4\times$
Forced Convection ($V = 2$ m/s)	24.7	1729	$43.7\times$

6 Comparison of Results

7 Parametric Analysis

7.1 Heat Transfer vs. Air Gap Thickness

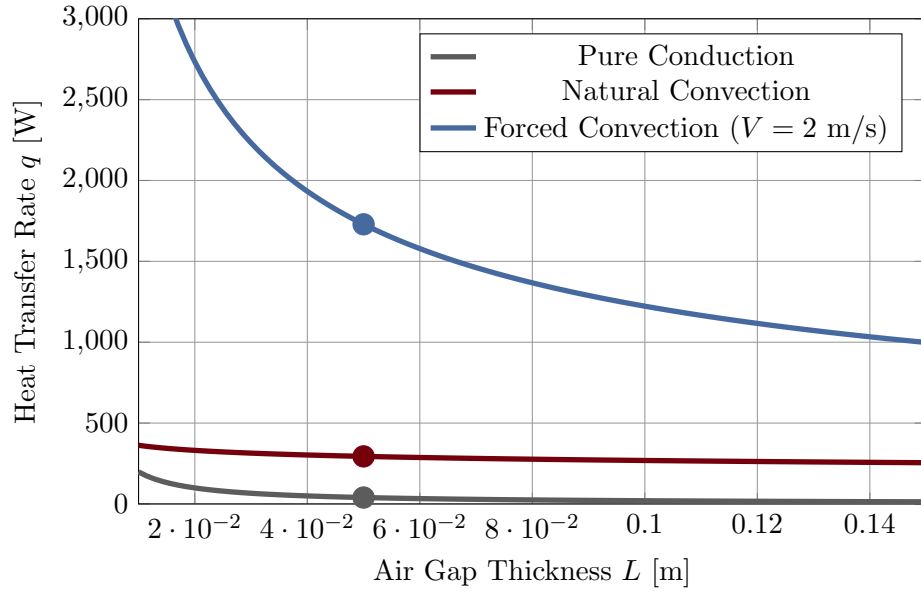


Figure 3: Heat transfer rate as a function of air gap thickness for different modes.

7.2 Heat Transfer Coefficient vs. Air Velocity (Forced Convection)

8 Conclusions

1. **Stagnant air** with natural convection provides $\approx 7\times$ enhancement over pure conduction due to buoyancy-driven circulation.
2. **Moving air** at $V = 2$ m/s provides $\approx 44\times$ enhancement over pure conduction and $\approx 6\times$ over natural convection.
3. The heat transfer coefficient scales as:
 - Conduction: $h \propto 1/L$
 - Natural convection: $h \propto L^{-0.13}$ (weak dependence)

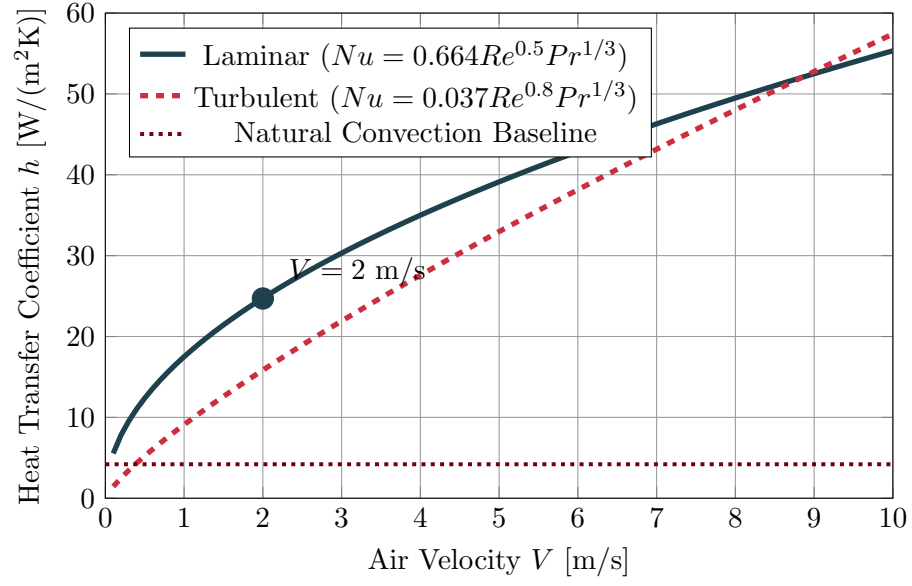


Figure 4: Heat transfer coefficient vs. air velocity for forced convection.

- Forced convection: $h \propto V^{0.5}$ (laminar) or $h \propto V^{0.8}$ (turbulent)
4. For maximum heat transfer, forced convection with high velocity and small gap thickness is preferred.